

Just Enough Process for AD

The concept of “good enough” software is getting more play in the industry. We discuss how to create consistency in AD delivery with “just enough” process.

Core Topic

Applications Development: Applications
Development Process Management

Key Issue

How will process management technology
evolve, and what will be the implications for
AD and non-AD organizations?

James Bach’s “The Challenge of Good Enough Software” asserts that there is a balance of market, quality and cost considerations that organizations must take into account when developing applications. The basic assertion is that the “zero defect” mentality, although appropriate as an ideal, is more than likely not the reality to which an applications development (AD) shop should aspire, especially those developing applications with business-driven, time-to-market constraints. We believe that the same assertion holds true with adherence to a software development process. In earlier research (see ADM *Strategic Analysis Report* R-480-127, Sept. 27, 1995), we proposed the AD Management Continuum — a model that specifies the critical components necessary for an AD shop’s development process and how to manage each iteration of that process (i.e., a project). Other models — which have varying levels of both complexity and coverage, such as the Software Engineering Institute’s Capability Maturity Model (CMM) — attempt to provide similar guidance. However, all such models function as ideals, rather than “cookbooks.” What are the key components that would indicate when an organization has “just enough” process to deliver software consistently?

AD Process Today: Although criticisms of the CMM and other models are often valid, using an abstraction (model) of “best in class” development processes is something every AD shop should do. The IT industry, as a whole, has been notoriously negligent in establishing standards, at least in the AD domain. However, much of this procrastination comes from intelligent, well-meaning individuals or groups disagreeing on the specifics of the model, rather than on the applicability of a model to the AD domain. Perhaps this is the result of the professional success of the critics, who have often been rewarded as a result of using their own internal (mental) models, rather than those available externally. However, while the “hacker” method of development

GartnerGroup

Entire contents © 1997 by Gartner Group, Inc. All rights reserved. Reproduction of this publication in any form without prior written permission is forbidden. The information contained herein has been obtained from sources believed to be reliable. Gartner Group disclaims all warranties as to the accuracy, completeness or adequacy of such information. Gartner Group shall have no liability for errors, omissions or inadequacies in the information contained herein or for interpretations thereof. The reader assumes sole responsibility for the selection of these materials to achieve its intended results. The opinions expressed herein are subject to change without notice.

may be effective for the extremely creative and intelligent (or those whose offices are replete with pinball machines and free caffeine), the remainder of the marketplace could use help.

Note 1

"Just Enough" Process for AD

Methodology

- A set of delivery routes, or templates, for the primary types of development done by the organization (e.g., a large AD shop might include routes for legacy maintenance, client/server development, package implementation and rapid applications development)
- Ability to use both internal and external best practices as guidelines

Process Management

- A tool to enable methodology delivery to the desktop of the AD constituencies: methodologist, manager and developer
- Ability to add routes from external sources
- Rudimentary metrics capabilities at a project level

Project Management

- Ability to download and customize development routes
- Defined, repeatable, second-order project management processes included in each route
- Project management toolset to create, schedule, level resources and report actuals

Just Enough Process: Although a specified dose of antibiotics fights infections, taking entire bottles of a drug will not cure the disease quicker; in fact, it can be very harmful. In our example, if a little process is good, is much more much better? AD shops must take a determined, pointed look at their development efforts and then establish the exact level of process and rigor that is crucial for delivery of software. The degree of standardization and enforcement will vary by organization — one size does not fit all when it comes to how much is "just enough." Note 1 suggests some recommendations for "just enough" process, using the AD Management Continuum categories as a template. However, this is not the only model that should be used. The aforementioned CMM — as well as process models from Software Productivity Research and Pressman & Associates — will serve equally well as a guide. The model chosen must be applied consistently. We do not make these suggestions as a means for process-averse shops to avoid rigor, nor as an excuse for loosening up an already followed process. The concept also has application when considering the constituencies doing software development. If "traditional," centralized IT development is at one end of a continuum and end-user development is at the other, there will be differences in the degree of process rigor necessary not only at the organization level, but at the development constituency level as well.

Not Enough Process: Though the degree of rigor of process will vary by organization (and, perhaps, within organization), the basic need for process will not. Our notion of "just enough" process may not be suitable for the "process neophyte" organization — one that must first learn the basics and follow them rigorously, then learn where, when and how to apply them at proper levels. Our recommendations and comments should not be seen as "permission" for an organization to abdicate its process model responsibilities. The risks of doing so are apparent in the success, or lack of success, in most software organizations and are simply unacceptable for the AD shop that is driven by the need to effectively fulfill its client's needs.

Bottom Line: It is a truism that every organization needs development processes and must follow them with a certain degree of rigor. However, process is not the name of the game — delivering software is. If too much rigor costs too much and takes too long, the business supported by AD will cease to exist. By employing "just enough" process, organizations can balance the need for rigor with the reality of existence in the AD shop — minimize cost and risk, maximize benefit for the business.

